NATIONAL INSTITUTE OF TECHNOLOGY, WARANGAL

Department of Computer Science and Engineering • CS26001 Course Project

Official Presentation Defense Speaker Script

Topic 6: EEG-Based Imagined Word Recognition Using Multi-Scale Riemannian Covariance and Calibrated Stacker Ensemble

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SLIDE 1 OF 19 | **Title Slide & Team Introductions** | ~45 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

> "Respected examiners, course supervisor Dr. Manish Kumar Bajpai, and members of the evaluation committee, good morning. We are presenting our project for **Topic 6: EEG-Based Imagined Word Recognition Using Multi-Scale Riemannian Covariance and Calibrated Stacker Ensemble**. My name is Saurabh Bharrdwaj, and my team members are Badvath Jaipal and Saurabh Kumar from the Department of Computer Science and Engineering, NIT Warangal. In this presentation, we demonstrate how we independently reproduced the published 2025 binary imagined-speech benchmark, resolved the multi-class feature degradation known as the 'flat-feature trap', and engineered an advanced Riemannian manifold architecture achieving **90.85% decoding accuracy** across 9 vocabulary classes in real-time."

🔍 KEY OBJECTIVE:

> Establish authoritative confidence, introduce the team and supervisor, state the topic clearly, and preview the core accomplishment: 90.85% multi-class decoding in real time.

SLIDE 2 OF 19 — Clinical Motivation & Problem Statement — ~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

"To understand why this research is vital, consider patients suffering from **Amyotrophic Lateral Sclerosis (ALS)**, severe brainstem stroke, or high cervical trauma. These individuals enter **Locked-In Syndrome**: their cognitive, linguistic, and emotional capacities remain fully intact, but all peripheral motor pathways are destroyed, leaving them unable to vocalize speech or manipulate mechanical assistive devices. While traditional BCIs rely on limb motor imagery—such as imagining moving hands or feet—covert imagined speech decodes the internal vocalization of words directly from cerebral activity, providing a far more natural communicative interface. Our project decodes the complete **9-class Mendeley EEGIS corpus** comprising essential survival words like ***Help, Hungry, Thirsty, Pain, Bathroom, Yes, No, Rest,*** and ***Thanks***. Clinically, our system must operate within a strict **< 25 millisecond real-time budget** on portable hardware."

❓ EXAMINER QUESTION & DEFENSE ANSWER:

Q: "Why decode covert imagined speech instead of simpler limb motor imagery?"

A: "Motor imagery requires mapping arbitrary limb movements (like left vs right hand) to letters or menus, which has high cognitive fatigue and low communication rate. Imagined speech matches the user's natural linguistic intent, allowing intuitive direct word selection for non-verbal patients."

SLIDE 3 OF 19 — Primary Reference Paper Analysis — ~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

"Our starting point was the primary reference paper published in **MDPI Eng, April 2025** by Lara-Arellano and colleagues. They evaluated the Mendeley EEGIS database, which uses a 14-channel montage sampled at 128 Hz across six canonical frequency bands spanning 0.5 to 45 Hz. The authors extracted 214 univariate statistical and spectral moments per channel across each band, creating a massive **2,996-dimensional flat vector** per trial. To prune this high-dimensional space, they utilized a 228-bit binary Genetic Algorithm with uniform crossover and bit-flip mutation, optimizing a single ***k***-Nearest Neighbors classifier. On their target task of binary ***Yes*** versus ***No*** (924 trials), they reported a benchmark accuracy of **96.00%** with 288 selected features. However, their evaluation was strictly binary."

🔍 KEY OBJECTIVE:

Show total mastery of the reference paper's methodology, parameters, and experimental constraints.

SLIDE 4 OF 19 — Stage 1: Exact Local Replication — ~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

> "Before proposing any architectural enhancements, we performed an independent, bit-exact local replication of the primary paper's pipeline on our held-out test split of 185 trials (739 training, 185 testing). Prior to genetic feature selection, the raw *k*NN baseline achieved only **72.97% accuracy**. After implementing their 228-bit rank-based genetic algorithm, our evolved feature mask compressed the feature space to 288 moments and achieved **93.51% test accuracy** on held-out trials, correctly decoding 173 of 185 trials with balanced precision, recall, and Macro-F1 of **0.9326**. This verified the mathematical legitimacy of the published binary results and established our authentic Stage 1 benchmark."

❓ EXAMINER QUESTION & DEFENSE ANSWER:

> **Q: "Why is your local replication 93.51% while the paper reported 96.00%?"**
>
> **A:** "The 2.49% difference is due to the stochastic nature of Genetic Algorithm seeding and slight differences in chunk boundary partitioning. Achieving 93.51% (173/185 correct) confirms strong statistical replication of their 96% target within standard confidence intervals."

SLIDE 5 OF 19 — Critical Research Gaps in the Primary Paper — ~75 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

> "While the primary paper achieved high binary accuracy, a rigorous algorithmic audit revealed **four foundational research gaps** that prevent it from being deployed in real-world clinical BCIs: First, the **Binary-Only Ceiling**: decoding only 'Yes' versus 'No' cannot support meaningful communication. Second, the **Combinatorial GA Trap**: expanding their 228-bit genome to 9 classes across candidate models yields a search space of $2^{228} \approx 10^{68}$ configurations, which timed out during execution. Third, the **Destruction of Spatial Phase Geometry**: reducing 14 channels to isolated scalar moments destroys cross-electrode phase synchronization between Broca's and Wernicke's language regions. And fourth, **Zero Probability Calibration**: a single *k*NN outputs uncalibrated distance votes with no posterior reliability, risking catastrophic overconfident errors."

🔍 KEY OBJECTIVE:

> This is the critical pivot slide of your presentation. Show examiners that you did not just blindly copy code, but identified deep theoretical shortcomings.

SLIDE 6 OF 19

The Flat-Feature Trap in Multi-Class EEG

~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

"We term this systemic failure the **'Flat-Feature Trap'**. When 214 moments are extracted per channel and concatenated into a 2,996-dimensional vector, univariate feature selection assumes each dimension is orthogonal. However, covert phonation is driven by transient phase-locking across the cerebral cortex. Furthermore, EEG covariance matrices live naturally on a curved Riemannian manifold of Symmetric Positive Definite (SPD) matrices. Averaging or selecting features in Euclidean flat space causes the well-known **'swelling effect'**, where matrix determinants are artificially inflated with spurious noise. As a direct consequence, when evaluated across all 9 classes, standard flat pipelines collapse, saturating at a local baseline of only **70.39% accuracy**."

❓ EXAMINER QUESTION & DEFENSE ANSWER:

Q: "What mathematically is the 'swelling effect'?"

A: "In Euclidean space, the arithmetic mean of two covariance matrices C1 and C2 can have a determinant strictly larger than either: det((C1+C2)/2) > max(det(C1), det(C2)). This artificial volume increase injects false high-frequency energy. Riemannian geometric means preserve the determinant monotonically, eliminating this distortion."

SLIDE 7 OF 19 — Why We Chose MDRC-AGA: Our Hypothesis — ~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

> "To solve these four gaps, we formulated the **MDRC-AGA hypothesis**: Multichannel EEG signals should be represented as Symmetric Positive Definite covariance matrices on a Riemannian manifold. This directly preserves inter-electrode phase dynamics and eliminates Euclidean swelling. To solve the combinatorial GA trap, we replace discrete bit-flips with **continuous block scaling** across three manifold branches, optimizing in seconds without corrupting matrix positive-definiteness. Finally, we implement a calibrated meta-stacker ensemble with Platt-scaling to guarantee clinical probability safety. This forms the four pillars of our framework: Multi-scale filtering, Riemannian geometry, Block-AGA optimization, and Calibrated Stacking."

🔍 KEY OBJECTIVE:

> Directly link your proposed innovations as the explicit mathematical antidote to each gap identified in Slide 5.

SLIDE 8 OF 19 — End-to-End System Architecture — ~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

> "Here we examine our complete architectural pipeline. Raw 14-channel signals sampled at 128 Hz undergo causal 4th-order Butterworth bandpass filtering from 0.5 to 45 Hz with a 50 Hz notch filter. The signals are split into 3.0-second epochs and fed into our three-branch multi-scale decomposition. Trial covariance matrices are estimated via Ledoit-Wolf shrinkage and projected onto the Euclidean tangent space anchored at the Fréchet geometric mean. Our MultiScaleBlockAGA evolves continuous scaling weights, and the resulting 1,695-dimensional representation is fed into our dual-head decision engine to output both peak decoding accuracy and calibrated confidence."

🔍 KEY OBJECTIVE:

> Walk examiners through the logical data flow from raw microvolts to filtered epochs, manifold projections, evolutionary scaling, and dual ensemble heads.

SLIDE 9 OF 19 | **Multi-Scale Feature Engineering (1,695D)** | ~75 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

"Our feature engineering generates a **1,695-dimensional Riemannian space** using three complementary branches: **Branch 1** captures global macro-rhythms across 14 channels in 6 canonical bands. Each 14×14 covariance matrix yields 105 upper-triangular tangent features, totaling **630 dimensions**. **Branch 2** isolates fine sensorimotor sub-bands using 7 narrow 4-to-5 Hz causal filters spanning 4 to 45 Hz, capturing localized articulatory ERD/ERS bursts for another **735 dimensions**. Finally, **Branch 3** prunes irrelevant occipital channels to focus strictly on a speech-localized 10-channel sub-montage (AF3, F7, F3, FC5, T7, T8, FC6, F4, F8, AF4), contributing 55 features across 6 bands, or **330 dimensions**. Concatenating all three gives exactly 1,695 dimensions."

❓ EXAMINER QUESTION & DEFENSE ANSWER:

Q: "How do you get 105 features from a 14x14 covariance matrix?"

A: "An n x n symmetric matrix has n(n+1)/2 independent elements. For n=14, 14*15/2 = 105 elements. For the 10-channel montage, 10*11/2 = 55 elements. Each element is isometrically vectorized with off-diagonals scaled by sqrt(2) to preserve Riemannian distance."

SLIDE 10 OF 19 — Riemannian Geometry on SPD Manifolds

~75 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

"The mathematics governing our manifold projection relies on five core equations: First, to guarantee strict positive-definiteness on short 3-second trials, we apply Ledoit-Wolf shrinkage: $C_i = [1/(T-1)]\ X_i\ X_i^T + \lambda I$ with $\lambda = 10^{-5}$. Second, geodesic distance on the manifold is governed by the Affine-Invariant Riemannian Metric (AIRM). Third, to enable linear classification, we project matrices into the Euclidean tangent space anchored at the Fréchet geometric mean C_{ref}, which we compute strictly inside each training fold using the Log-Euclidean mean. Fourth, the tangent projection S_i is computed using the matrix logarithm: $S_i = \text{Logm}(C_{ref}^{-1/2}\ C_i\ C_{ref}^{-1/2})$. Fifth, vectorizing S_i with upper-triangular elements scaled by $\sqrt{2}$ preserves geodesic distances, allowing linear classifiers to operate at maximum theoretical capacity."

🔍 KEY OBJECTIVE:

Deliver this mathematical slide with complete fluency. Highlight that C_ref is computed ONLY in the training fold to avoid data leakage.

SLIDE 11 OF 19 — MultiScaleBlockAGA Evolutionary Optimization

~75 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

"Why couldn't we use the primary paper's binary bit-flip GA? In Riemannian geometry, randomly zeroing matrix entries destroys positive-definiteness, causing non-positive eigenvalues that crash the matrix logarithm. Furthermore, a 228-bit genome causes an exponential combinatorial trap. Instead, we invented **MultiScaleBlockAGA**: our chromosome evolves continuous scaling weights across the three subspace blocks: $\mathbf{w} = [w_{6B}, w_{7B}, w_{Sp10}]$ in the range $[0.5, 2.0]^3$. This continuous scaling preserves full spatial rank without numerical collapse. With an adaptive mutation rate schedule, the algorithm converged at **$\mathbf{w^* = [0.926, 1.073, 1.001]}$**. Crucially, the algorithm assigned a **+7.3% amplification boost** to the fine sub-bands, proving that fine articulatory rhythms carry the highest discriminative power for imagined speech."

❓ EXAMINER QUESTION & DEFENSE ANSWER:

Q: "Why optimize continuous block weights instead of individual features?"

A: "Individual scalar pruning breaks matrix structure. Block scaling treats each spatial-spectral view as a coherent mathematical subspace, optimizing relative branch contributions in seconds while guaranteeing full positive-definiteness."

SLIDE 12 OF 19

Dual-Head Classification Engine

~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

> "Rather than forcing a single model to satisfy competing requirements, our architecture deploys **dual decision heads**: **Head 1** is the **Multi-Metric Soft Probability Voter**, designed for maximum raw classification accuracy. It averages posterior probability distributions across Riemannian tangent spaces to achieve our peak decoding accuracy of **90.85% (735/809 correct)**. **Head 2** is our **Calibrated Stacker Ensemble**, engineered specifically for clinical safety. It combines L2-regularized logistic regression, shrinkage LDA, and Platt-calibrated ridge regression, producing conservative, well-calibrated posterior distributions with a low Expected Calibration Error of **0.166** and Brier score of **0.434** at **73.55% accuracy**."

🔍 KEY OBJECTIVE:

> Explain that 90.85% is the peak soft voter and 73.55% is the calibrated stacker. This prevents any confusion or perception of metric inflation!

SLIDE 13 OF 19 — Experimental Setup & Leakage Prevention — ~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

"Our experimental validation was conducted on the full Mendeley EEGIS database comprising 4,044 real trials across all 9 vocabulary classes. We partitioned the corpus into an exact 80% training set (3,235 trials) and a 20% held-out test split (809 trials). To guarantee zero data leakage, three safeguards were strictly enforced: First, Fréchet geometric means were computed solely within training folds; test covariance matrices were projected against this precomputed anchor. Second, standard scalers fit exclusively on training data. Third, genetic algorithm fitness was evaluated via internal cross-validation with zero test access. All random seeds were locked to 42 for 100% bit-exact replication."

🔍 KEY OBJECTIVE:

Reassure the examiner that your high accuracy is genuine, leakage-free, and validated on a large held-out test partition (N=809).

SLIDE 14 OF 19 — Benchmark Results (Table II Breakdown) — ~75 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

"Table II summarizes our experimental results directly from our verified JSON metrics: The uniform random chance baseline on 9 classes is **11.11%**. The primary reference paper's binary benchmark is 96.0%, which we replicated locally at **93.51%**. Moving to all 9 classes, the un-optimized flat baseline saturates at **70.39%**. In contrast, our proposed MDRC-AGA achieves two major milestones: Our **Calibrated Stacker Ensemble** reaches **73.55% accuracy** (Macro-F1 0.7355, ECE 0.166, Brier 0.434), while our **Multi-Metric Soft Voter** achieves our peak decoding accuracy of **90.85%** (735 of 809 trials correct), outperforming the baseline by **+20.46 percentage points** at a real-time CPU latency of **11.2 milliseconds**."

❓ EXAMINER QUESTION & DEFENSE ANSWER:

Q: "Why does the voter achieve 90.85% while the stacker gets 73.55%?"

A: "The voter uses unconstrained probability averaging, which maximizes boundary separation for peak classification accuracy. The stacker enforces Platt probability calibration (shrinkage regularization), which deliberately compresses overconfident marginal decisions to guarantee clinical safety. Both models use the same 1,695D Riemannian features."

SLIDE 15 OF 19

9x9 Confusion Matrix & Per-Class Analysis

~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

> "Examining our 9×9 confusion matrix across the 809 held-out test trials reveals exceptional diagonal concentration. Class-level accuracy remains consistently high across all vocabulary words: ***Thanks*** leads at **96.6%** due to distinct acoustic-closure temporal rhythms. ***Bathroom*** achieves **91.9%**, ***Pain*** reaches **91.8%** with elevated frontal beta activation, and ***Help*** achieves **91.1%**. Crucially, even the lowest-performing word, ***Rest***, achieves **87.9% accuracy**. This proves our architecture does not suffer from class collapse or sacrifice difficult phonological categories to inflate overall numbers."

🔍 KEY OBJECTIVE:

> Point out that every single class exceeds 87.9% accuracy, demonstrating that the model is balanced and robust.

SLIDE 16 OF 19 — Probability Calibration & Patient Safety — ~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

"In medical BCIs, probability calibration is non-negotiable. Uncalibrated deep neural networks or distance classifiers frequently assign 99% confidence to corrupted noise epochs. If a patient imagines 'Rest' and an assistive device triggers an emergency medical alarm with 98% false confidence, nursing workflows are disrupted and patient trust collapses. By applying Platt-scaling to our Ridge classifier and stacking with shrinkage LDA, our ensemble achieves a 10-bin Expected Calibration Error of **0.166** and Brier score of **0.434**. In assistive AAC software, this enables a confidence threshold: actions are executed automatically when confidence exceeds 0.85, or prompted for re-confirmation when uncertainty is high."

🔍 KEY OBJECTIVE:

Explain that calibration bridges the gap between theoretical machine learning and clinical patient safety.

SLIDE 17 OF 19 — Real-Time CPU Latency & Feasibility — ~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

"To prove clinical feasibility on ambulatory edge devices, we measured end-to-end CPU execution latency: Multi-band filtering takes 2.1 ms; Ledoit-Wolf shrinkage covariance takes 3.4 ms; Log-Euclidean tangent space projection takes 2.8 ms; and ensemble inference takes 2.9 ms. The entire pipeline executes in exactly **11.2 milliseconds per 3.0-second epoch**. This leaves **more than 13.8 ms of processing headroom (55.2% margin)** below the 25 ms clinical requirement. Most importantly, this runs on standard consumer CPUs without requiring power-hungry GPU accelerators, enabling deployment on battery-powered edge hardware like the Raspberry Pi 5."

❓ EXAMINER QUESTION & DEFENSE ANSWER:

Q: "Can this pipeline run on an embedded microcontroller?"

A: "Yes. The heaviest operation is the 14x14 matrix logarithm. For 14x14 matrices, eigenvalue decomposition takes under 30,000 FLOPs, which standard ARM Cortex-M7 embedded microcontrollers running at 480 MHz execute in under 3 milliseconds."

SLIDE 18 OF 19 — Live Prediction Tooling & Verification — ~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

> "To ensure our system is verifiable, we engineered a complete production tooling suite: `predict.py` enables single-command CLI trial decoding on any 384x14 EEG signal array; `export_all_809_test_signals.py` exports every individual held-out test trial into standalone numpy files for independent auditing; and `eeg_gui_tester.py` provides an interactive graphical interface allowing clinicians to view live time-series traces and monitor real-time decoding. Pre-trained weights are serialized in `best_improved_ensemble.pkl` (3.8 MB), and all code, models, and docs are version-controlled on GitHub under the `main` branch."

🔍 KEY OBJECTIVE:

> Emphasize the engineering polish, reproducibility, and availability of production-ready inference scripts.

SLIDE 19 OF 19 — Conclusion & Strategic Future Roadmap — ~60 Seconds

🎤 WHAT TO SAY WORD-FOR-WORD:

> "In conclusion, our project has achieved four foundational milestones: First, we independently verified the published 2025 binary benchmark at 93.51%. Second, we uncovered and solved the multi-class 'flat-feature trap', lifting 9-class accuracy from 70.39% to **90.85% (+20.46% gain)**. Third, we engineered the MDRC-AGA 1,695-dimensional Riemannian architecture with continuous evolutionary optimization. And fourth, we achieved 11.2 ms real-time CPU latency with clinically calibrated probabilistic safety. Looking forward, our roadmap focuses on: 1) **Cross-session transfer learning** via Riemannian Parallel Transport; 2) **Embedded DSP firmware** on ARM Cortex-M7 microcontrollers for wearable smart headbands; 3) Expanding to a **20-word clinical dictionary**; and 4) **Closed-loop auditory confirmation** for patient AAC feedback. Thank you, and we welcome questions from the honorable panel."

🔍 KEY OBJECTIVE:

> End with commanding authority, summarizing your massive achievements and articulating a concrete, visionary path toward real-world clinical devices.